

DOCUMENT #8 OF 10 — SYSTEM SPECIFICATION

Software Requirement Specification (SRS)

Detailed Functional Scope, User Capabilities, RBAC Access Control & System Edge Cases

Compliance Standard	Security Baseline	System Scope	Document Version
IEEE 830 / ISO-IEC 25010	PostgreSQL RLS Multi-Tenant	AI Resume Screening & Assessment	v1.0 (Production SRS)

1. Role-Based Access Control (RBAC) Matrix

The system strictly isolates data access using PostgreSQL Row Level Security (RLS) linked to Supabase Auth roles.

User Action / Module	Super Admin	Recruiter / HR	Candidate
Create / Edit Jobs & JDs	Full Access	Own Organization Only	No Access
Upload Bulk Resumes	Full Access	Own Projects	Upload Own Resume Only
View Global Candidate Ranking	Full Access	Own Job Screenings	No Access
Candidate Triage (Shortlist/Reject)	Full Access	Allowed	No Access
AI Interview & Voice Assessment	Audit / View Transcripts	View Results & Score	Take Assigned Interview
Billing & Organization Settings	Full Control	Org Admin Level	No Access

2. Detailed Functional Requirements (What Exactly Happens in App)

FR-1: Job Description Parsing & Taxonomy Construction

System Workflow:

- Recruiter submits job description text (min 20, max 2000 characters).
- Edge Function invokes Google Gemini / Claude API to generate a structured JSON entity taxonomy: mandatory_skills, optional_skills, min_experience_years, and education_level.
- Taxonomy saved to tasks table and rendered as selectable tags in the UI.

FR-2: Multi-Format Resume Ingestion & ATS Compatibility Linting

System Workflow:

- Candidate/Recruiter uploads PDF/DOCX files (≤ 5MB per file).
- Client-side validator checks MIME types and pushes raw binary to Supabase Storage private bucket (resumes-vault).
- ATS Linting evaluates text parseability, standard section headings (Education, Experience, Skills), column complexity, and issues an ATS score (0–100).

FR-3: Semantic Match, Scoring & Ranking Calculation

System Workflow:

- Extracted candidate profile vector and Job Description vector are compared using semantic cosine similarity.
- Deterministic skill match algorithm evaluates exact and synonym overlaps:
$$\text{Skill Match \%} = \left(\frac{\text{Matched Mandatory Skills}}{\text{Total Mandatory Skills}} \right) \times 100$$
- Composite score weights: Skills Match (50%), Experience Relevance (30%), Education Alignment (20%).
- Leaderboard dynamically updates and candidate rank order (1 to N) is persisted in candidates table.

FR-4: AI Virtual Interview & Browser Voice Transcription

System Workflow:

- Candidate initiates interview round → AI Interviewer generates 3–5 personalized technical questions based on resume gaps.
- Browser native Web Speech API listens to candidate microphone, transcribes voice to text in real-time, and streams candidate responses back to the serverless AI evaluator.
- Automated scoring rubric generates technical depth rating, confidence score, and summarized transcript.

3. Comprehensive Edge Cases & System Handlers

Edge Case Scenario	System Detection & State	Handled Resolution / Behavior
Scanned / Image-only PDF	PyMuPDF / pdfplumber returns 0 bytes of selectable text.	Triggers fallback OCR engine or prompts user: "Document is image-based, please upload a text PDF".
Duplicate Resume Upload	MinHash / Email check matches existing entry in database (>90% similarity).	Prompts recruiter: "Candidate Rahul Sharma already parsed for this job. Update existing record or skip?"
LLM API Rate Limit / Downtime	Gemini/Claude API returns 429 Rate Limit or 503 Service Unavailable.	Exponential backoff retry queue (max 3 retries) followed by fallback cache scoring model.
Microphone Access Denied	Browser rejects navigator.mediaDevices.getUserMedia permission.	Displays fallback notification enabling standard keyboard text input for the AI interview.
Network Drop During Upload	Client detects navigator.onLine === false mid-upload batch.	Upload paused; resumable chunk state cached in localStorage; auto-resumes upon reconnection.